

Basic Cisco Commands

By Marcus Nielson (2014)

Configuring Basic Switch Settings (Switch Examples)

Enter **enable** if the prompt has changed back to Switch>.

```
Switch> enable
Switch#
```

Enter global configuration mode.

```
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
The prompt changed again to reflect global configuration mode.
```

Assign the switch hostname.

```
Switch(config)# hostname S1
S1(config)#
```

Configure password encryption.

```
S1(config)# service password-encryption
S1(config)#
```

Assign class as the secret password for privileged EXEC mode access.

```
S1(config)# enable secret class
S1(config)#
```

Prevent unwanted DNS lookups.

```
S1(config)# no ip domain-lookup
S1(config)#
```

Configure a MOTD banner.

```
S1(config)# banner motd #
Enter Text message. End with the character '#'.
Unauthorized access is strictly prohibited. #
```

Verify your access settings by moving between modes.

```
S1(config)# exit
S1#
*Mar 1 00:19:19.490: %SYS-5-CONFIG_I: Configured from console by console
S1# exit
```

S1 con0 is now available.

Creating/Assigning a VLAN IP Address Example:

```
S1(config)# vlan 99  
S1(config-vlan)# name (name of VLAN)  
S1(config-vlan)# exit  
S1(config)#
```

Configure the VLAN 99 IP address:

```
S1(config)# interface vlan 99  
S1(config-if)# ip address 172.16.99.11 255.255.255.0  
S1(config-if)# no shutdown  
S1(config-if)# end  
S1#
```

Configuring IP Default Gateway:

```
S1(config)# ip default-gateway 192.168.1.1  
S1(config)#
```

Restrict Access:

```
S1(config)# line con 0  
S1(config-line)# password (pw)  
S1(config-line)# login  
S1(config-line)# logging synchronous  
S1(config-line)# exit  
S1(config)#
```

VTY:

```
S1(config)# line vty 0 15  
S1(config-line)# password (pw)  
S1(config-line)# login  
S1(config-line)# end  
S1#
```

Assign ports F0/5 and F0/6 to VLAN 99 on the switch:

```
S1# config t  
S1(config)# interface f0/5  
S1(config-if)# switchport mode access  
S1(config-if)# switchport access vlan 99  
S1(config-if)# interface f0/6  
S1(config-if)# switchport mode access  
S1(config-if)# switchport access vlan 99  
S1(config-if)# end
```

Assigning ports to VLAN (range):

```
S1(config)# interface range f0/1 – 24,g0/1 - 2  
S1(config-if-range)# switchport access vlan #  
S1(config-if-range)# exit  
S1(config)# end
```

Configure and Verify SSH:

```
S1(config)# line vty 0 15  
S1(config-line)# transport input ssh  
S1(config-line)# login local  
S1(config-line)# exit
```

Generate an RSA crypto keys:

```
S1(config)# crypto key generate rsa modulus 1024
```

Configure general security features:

```
S1(config)# interface range f0/1 – 4  
S1(config-if-range)# shutdown  
S1(config-if-range)# interface range f0/7 – 24  
S1(config-if-range)# shutdown  
S1(config-if-range)# interface range g0/1 – 2  
S1(config-if-range)# shutdown  
S1(config-if-range)# end  
S1#
```

Enter local passwords:

```
S1(config)# enable secret class  
S1(config)# line con 0  
S1(config-line)# password (pw)  
S1(config-line)# login  
S1(config-line)# exit  
S1(config)#
```

Configure a trunk port:

```
S2(config)# interface f0/1  
S2(config-if)# switchport mode trunk  
S1# show interface trunk
```

Configuring 802.1Q Trunk-Based Inter-VLAN Routing:

```
S1(config)# vlan 10  
S1(config-vlan)# name Students  
S1(config-vlan)# vlan 20  
S1(config-vlan)# name Faculty  
S1(config-vlan)# exit  
S1(config)# interface f0/1  
S1(config-if)# switchport mode trunk
```

```
S1(config-if)# interface f0/5
S1(config-if)# switchport mode trunk
S1(config-if)# interface f0/6
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
```

Configure DHCPv4 :

```
S1(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
S1(config)# ip dhcp pool DHCP1
S1(dhcp-config)# network 192.168.1.0 255.255.255.0
S1(dhcp-config)# default-router 192.168.1.1
S1(dhcp-config)# dns-server 192.168.1.9
S1(dhcp-config)# lease 3
```

Configure DHCPv4 for Multiple VLANs:

```
S1(config)# interface f0/6
S1(config-if)# switchport access vlan 2
S1(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.10
S1(config)# ip dhcp pool DHCP2
S1(dhcp-config)# network 192.168.2.0 255.255.255.0
S1(dhcp-config)# default-router 192.168.2.1
S1(dhcp-config)# dns-server 192.168.2.9
S1(dhcp-config)# lease 3
```

Enable IP Routing:

```
S1(config)# ip routing
S1(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.10
R1(config)# ip route 192.168.2.0 255.255.255.0 g0/1
```

Enable spanning-tree, rapid-pvst, portfast:

```
S1(config)# spanning-tree vlan 1,10,99 root secondary
S1(config)# spanning-tree mode rapid-pvst
S1(config)# interface f0/6
S1(config-if)# spanning-tree portfast
S1(config-if)# spanning-tree bpduguard enable
```

Configure PAgP:

```
S1(config)# interface range f0/3-4
S1(config-if-range)# channel-group 1 mode desirable
Creating a port-channel interface Port-channel 1
S1(config-if-range)# no shutdown
```

Configure trunk ports assign to VLAN:

```
S1(config)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk native vlan 99
```

Delete the VLAN database:

```
S1# delete vlan.dat
Delete filename [vlan.dat]?
Delete flash:/vlan.dat? [confirm]
S1#
```

Configuring Basic Router Settings (Router Examples) =====

Configure the router:

```
Router> enable
Router#
Router# config terminal
Router(config)#
Router(config)# hostname R1.
R1(config)# no ip domain-lookup
R1(config)# security passwords min-length 10
R1(config)# enable secret cisco12345
```

Setting a console password on a router:

```
R1(config)# line con 0
R1(config-line)# password ciscoconpass
R1(config-line)# exec-timeout 5 0
R1(config-line)# login
R1(config-line)# logging synchronous
R1(config-line)# exit
R1(config)#
```

Assign VTY password:

```
R1(config)# line vty 0 4
R1(config-line)# password ciscovtypass
R1(config-line)# exec-timeout 5 0
R1(config-line)# login
R1(config-line)# logging synchronous
R1(config-line)# exit
R1(config)#
```

Encrypt the clear text passwords:

```
R1(config)# service password-encryption
```

Create a MOTD banner:

```
R1(config)# banner motd #Unauthorized access prohibited!#
```

Configure an IP address and interface description. Activate both interfaces on the router:

```
R1(config)# int g0/0
R1(config-if)# description Connection to PC-B
R1(config-if)# ip address 192.168.0.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# int g0/1
R1(config-if)# description Connection to S1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

Configure the router for SSH access:

```
R1# configure terminal
R1(config)# ip domain-name CCNA-lab.com
R1(config)# username admin privilege 15 secret adminpass1
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
R1(config-line)# exit
R1(config)# crypto key generate rsa modulus 1024
R1(config)# exit
```

Configure IPv6:

```
R1# configure terminal
R1(config)# interface g0/0
R1(config-if)# ipv6 address 2001:db8:acad:a::1/64
R1(config-if)# ipv6 address fe80::1 link-local
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# ipv6 unicast-routing
R1(config)# exit
```

Configure Trunk-Based Inter-VLAN Routing:

```
R1(config)# interface g0/1.1
R1(config-subif)# encapsulation dot1Q 1
R1(config-subif)# ip address 192.168.1.1 255.255.255.0
R1(config-subif)# interface g0/1.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# interface g0/1.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# exit
R1(config)# interface g0/1
```

```
R1(config-if)# no shutdown
```

Configure IP settings on the routers:

```
R3(config)# interface s0/0/0  
R3(config-if)# ip address 10.1.1.2 255.255.255.252  
R3(config-if)# clock rate 128000  
R3(config-if)# no shutdown
```

Configure RIPv2 routing:

```
R1# config t  
R1(config)# router rip  
R1(config-router)# version 2  
R1(config-router)# passive-interface g0/1  
R1(config-router)# network 172.30.0.0  
R1(config-router)# network 10.0.0.0
```

Disable automatic summarization:

```
R1(config)# router rip  
R1(config-router)# no auto-summary
```

Configure RIPv6 routing:

```
R1(config)# interface g0/1  
R1(config)# ipv6 rip Test1 enable  
R1(config)# interface s0/0/0  
R1(config)# ipv6 rip Test1 enable
```

Configure and Verify OSPF Routing:

```
R1(config)# router ospf 1  
  
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0  
R1(config-router)# network 192.168.12.0 0.0.0.3 area 0  
R1(config-router)# network 192.168.13.0 0.0.0.3 area 0
```

Configure IPv6 Addresses Manually:

```
R1(config)# interface g0/0  
R1(config-if)# ipv6 address 2001:db8:acad:a::1/64  
R1(config-if)# no shutdown  
R1(config-if)# interface g0/1  
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64  
R1(config-if)# no shutdown  
R1(config-if)# end  
R1#
```

Configure OSPFv6:

```
R1(config)# interface g0/0
```

```
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)# interface s0/0/0
R1(config-if)# ipv6 ospf 1 area 0
R1(config-if)# interface s0/0/1
R1(config-if)# ipv6 ospf 1 area 0
```

Configure a numbered standard ACL:

```
R3(config)# access-list 1 remark Allow R1 LANs Access
R3(config)# access-list 1 permit 192.168.10.0 0.0.0.255
R3(config)# access-list 1 permit 192.168.20.0 0.0.0.255
R3(config)# access-list 1 deny any
R3(config)# interface g0/1
R3(config-if)# ip access-group 1 out
```

Configure a named extended ACL:

```
R3(config)# ip access-list extended WEB-POLICY
R3(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 host 10.1.1.1 eq 80
R3(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 209.165.200.224 0.0.0.31 eq 80
R3(config-ext-nacl)# interface S0/0/1
R3(config-if)# ip access-group WEB-POLICY out
```

Configuring Basic DHCPv4:

```
R1(config)# router eigrp 1

R1(config-router)# network 192.168.0.0 0.0.0.255
R1(config-router)# network 192.168.1.0 0.0.0.255
R1(config-router)# network 192.168.2.252 0.0.0.3
R1(config-router)# no auto-summary
```

DHCP Configuration:

```
R2(config)# ip dhcp excluded-address 192.168.0.1 192.168.0.9
R2(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.9
R2(config)# ip dhcp pool R1G1
R2(dhcp-config)# network 192.168.1.0 255.255.255.0
R2(dhcp-config)# default-router 192.168.1.1
```

Configure HSRP:

```
R1(config)# interface g0/1
R1(config-if)# standby 1 ip 192.168.1.254
R1(config-if)# standby 1 priority 150
R1(config-if)# standby 1 preempt
```

Configure GLBP:

```
R1(config)# interface g0/1
R1(config-if)# glbp 1 ip 192.168.1.254
```

```
R1(config-if)# glbp 1 preempt
R1(config-if)# glbp 1 priority 150
R1(config-if)# glbp 1 load-balancing round-robin
```

Configure the ISP Router:

```
ISP(config)# username Cust1 password ciscoppoe
ISP(config)# ip local pool PPPoEPOOL 10.0.0.1 10.0.0.10
ISP(config)# interface virtual-template 1
ISP(config-if)# ip address 10.0.0.254 255.255.255.0
ISP(config-if)# mtu 1492
ISP(config-if)# peer default ip address pool PPPoEPOOL
ISP(config-if)# ppp authentication chap callin
ISP(config-if)# exit
```

Configure G0/0 interface as passive:

```
R1(config)# router eigrp 10
R1(config-router)# passive-interface g0/0
```

Configure EIGRP for IPv6 Routing

```
R1(config)# ipv6 router eigrp 1
R1(config-rtr)# router-id 1.1.1.1
```

Configure an SNMP agent:

```
R1(config)# snmp-server community cicolab ro SNMP_ACL
R1(config)# snmp-server location snmp_manager
R1(config)# snmp-server contact cicolab_admin
R1(config)# snmp-server host 192.168.1.3 version 2c cicolab
R1(config)# snmp-server enable traps
R1(config)# ip access-list standard SNMP_ACL
R1(config-std-nacl)# permit 192.168.1.3
```